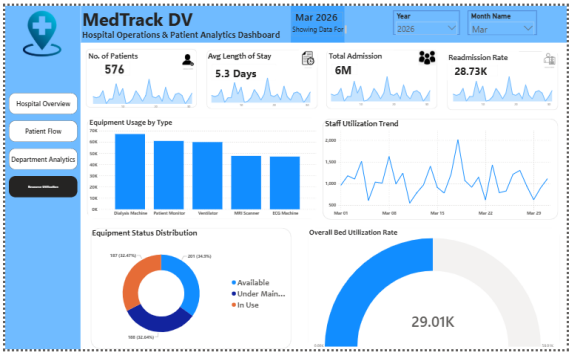
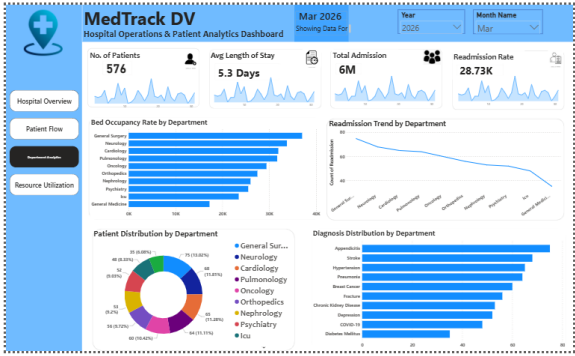
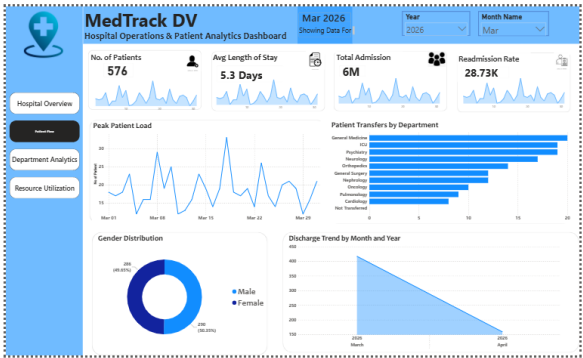
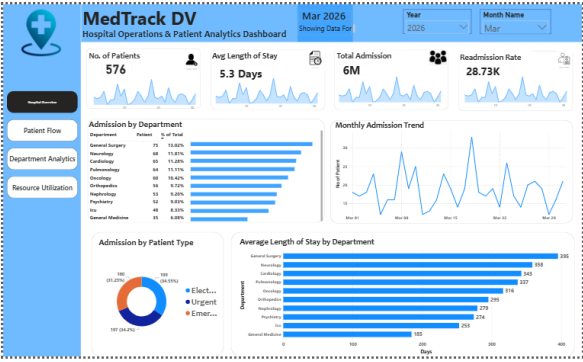


# MedTrack DV

## Hospital Operations & Patient Analytics Dashboard

Final Project Documentation

Built using Power BI Desktop



Hospital Overview · Patient Flow · Department Analytics · Resource Utilization

# 1. Introduction

## 1.1 Project Statement

This project aims to develop a comprehensive hospital operations and patient analytics dashboard suite for analyzing hospital performance, patient admissions, department efficiency, and healthcare resource utilization.

Using publicly available Hospital Datasets and Patient Admission Datasets, the project will transform healthcare operational data into actionable insights through interactive Power BI dashboards.

The dashboard suite will help hospital administrators, healthcare managers, medical staff, and policymakers monitor patient flow, evaluate department performance, optimize hospital resources, and improve operational efficiency through data-driven decision-making.

**The final deliverable is a unified Power BI file (.pbix) consisting of four interconnected dashboards:**

- Hospital Overview
- Patient Flow
- Department Analytics
- Resource Utilization

## 1.2 Objectives

The main objectives of this project are:

- To clean and preprocess hospital data using Python by handling missing values, duplicates, inconsistent data, and incorrect data formats.
- To analyze hospital operations and identify important patterns related to patients, admissions, departments, staff, beds, and other hospital resources.
- To develop meaningful Key Performance Indicators (KPIs) such as Total Admissions, Occupancy Rate, Average Length of Stay, Readmission Rate, Bed Utilization Rate, and Department Efficiency Score.
- To create interactive Power BI dashboards for Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization.
- To provide interactive data exploration using slicers, filters, and navigation controls.
- To identify department-level and hospital-level insights that can help in monitoring performance and resource utilization.
- To present complex hospital data through clear visualizations that make the information easier to understand and support data-driven decision-making.
- To validate the dashboards and their functionality through testing and a Quality Assurance (QA) checklist.

## 1.3 Workflow Diagram

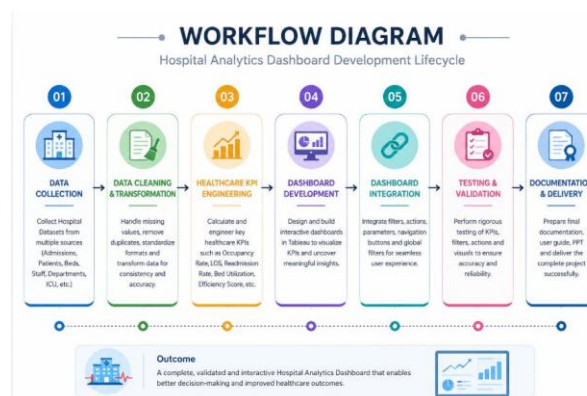


Figure 1: Hospital Analytics Dashboard Development Lifecycle

## 2. Data Collection, Cleaning & Transformation

---

### 2.1 Dataset Overview

The project uses a hospital healthcare dataset containing patient-level information related to admissions, hospital departments, beds, staff, transfers, and patient outcomes. The dataset was used to develop healthcare KPIs and interactive Power BI dashboards for Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization.

The dataset was initially provided in Excel format and was processed using Python/Pandas before being used for dashboard development.

### 2.2 Data Source

The primary dataset used for the project was:

**Dataset Name:** Hospital\_RawDataset\_Updated

**Working File:** <https://www.kaggle.com/datasets/hassanjameelahmed/healthcare-analytics-patient-flow-data>

The dataset contains hospital and patient-related records that support analysis of patient flow, hospital capacity, departmental performance, and resource utilization.

### 2.3 Dataset Size

The initial dataset contained:

- Rows: 10,000 patient records
- Columns: Multiple patient, hospital, department, staff, bed, admission, discharge, and transfer-related attributes
- Primary/Common Key: Patient\_ID

Each row represents a hospital-related patient record identified using Patient\_ID.

### 2.4 Columns

The dataset contains features belonging to several analytical categories:

| Category               | Example Features                            | Purpose                                      |
|------------------------|---------------------------------------------|----------------------------------------------|
| Patient Information    | Patient_ID, Patient-related attributes      | Identifies and describes patients            |
| Hospital Information   | Hospital Name, Hospital Type, State, City   | Enables hospital-level analysis              |
| Admission Information  | Admission Date, Admission Type              | Analyzes patient admissions                  |
| Discharge Information  | Discharge Date                              | Analyzes patient discharge and stay duration |
| Patient Flow           | Transfer Date, Transfer-related information | Analyzes patient transfers                   |
| Department Information | Department                                  | Enables department-wise analysis             |
| Bed Information        | Beds Occupied, ICU Beds                     | Measures hospital capacity and utilization   |
| Staff Information      | Staff Count                                 | Measures staff availability and utilization  |

| Category       | Example Features | Purpose                                      |
|----------------|------------------|----------------------------------------------|
| Length of Stay | Length of Stay   | Measures duration of patient hospitalization |

## 2.5 Data Cleaning and Transformation

### 2.5.1 Missing Values

Missing-value analysis was performed for all columns.

The most significant missing-value issue was identified in the Transfer\_Date column. A large number of records did not contain a transfer date because the corresponding patients were not transferred.

Observed missing values in Transfer\_Date: 8,240 records.

Since a missing transfer date could represent a patient who was never transferred, these values were handled by replacing them with: "Not Transferred". This makes the information easier to interpret during dashboard analysis rather than treating every missing value as an unknown value.

### 2.5.2 Duplicate Records

Duplicate records were checked using Pandas.

Duplicate records identified: 0

Therefore, no duplicate patient records required removal.

### 2.5.3 Incorrect Data Types

The data types of the columns were reviewed and corrected where necessary. Particular attention was given to:

- Date columns
- Numerical columns
- Categorical/text columns
- Patient identifiers

Date fields were converted into appropriate datetime formats, while numerical measures were converted into suitable numeric data types.

### 2.5.4 Date Corrections

Date fields were converted into a standardized date format. The following types of dates were examined:

- Admission Date
- Discharge Date
- Transfer Date

Date validation was also performed to identify logically invalid records.

### 2.5.5 Length-of-Stay Issues

The Length of Stay field was compared with the difference between admission and discharge dates.

A mismatch was identified in 6,051 records between the existing Length of Stay value and the value derived from the admission and discharge dates. These records were identified as part of the data-quality validation process and were reviewed before using Length of Stay for KPI calculations.

Additionally, 2,964 records were identified where the discharge date occurred before the admission date. These records represent logically inconsistent date relationships and were flagged during the cleaning process.

This validation was important because incorrect Length of Stay values could directly affect the Average Length of Stay KPI and patient-flow analysis.

2.6 Data Quality Summary

| Data Quality Check         | Result                     | Action Taken                             |
|----------------------------|----------------------------|------------------------------------------|
| Initial Records            | 10,000                     | Retained for analysis                    |
| Duplicate Records          | 0                          | No removal required                      |
| Missing Transfer Dates     | 8,240                      | Replaced with Not Transferred            |
| Length-of-Stay Mismatches  | 6,051                      | Identified and validated                 |
| Discharge Before Admission | 2,964                      | Identified as invalid date relationships |
| Date Formatting            | Inconsistent               | Standardized                             |
| Data Types                 | Required correction        | Converted to appropriate types           |
| Categorical Values         | Inconsistencies identified | Standardized                             |
| Final Dataset              | Cleaned dataset            | Used for KPI and dashboard development   |

3. KPI Engineering

3.1 Overview

Healthcare KPI Engineering is the process of converting the cleaned hospital dataset into meaningful performance indicators. These KPIs provide quantitative measures of hospital operations, patient flow, departmental performance, and resource utilization.

The calculated KPIs were designed to support the four project dashboards:

- Hospital Overview
- Patient Flow
- Department Analytics
- Resource Utilization

The KPIs were implemented using DAX measures and aggregation logic in Power BI.

3.2 KPI Definitions and Formulas

3.2.1 Total Admissions

**Definition:** Total number of patient admission records during the selected period.

**Formula:** Total Admissions = COUNT(Patient\_ID)

**Purpose:** Measures the overall volume of patients admitted to the hospital.

**Measurement:** Count

3.2.2 Occupancy Rate

**Definition:** Percentage of available hospital beds occupied by patients.

**Formula:**  $\text{Occupancy Rate} = \text{AVERAGE}([\text{Bed Occupancy Rate Calc}])$

**Purpose:** Helps determine hospital capacity utilization.

**Measurement:** Percentage

### 3.2.3 Average Length of Stay

**Definition:** Average number of days patients remain hospitalized.

**Formula:**  $\text{Average Length of Stay} = \text{AVERAGE}(\text{DATEDIFF}([\text{Admission Date}], [\text{Discharge Date}], \text{DAY}))$

**Purpose:** Helps evaluate patient flow and hospital resource usage.

**Measurement:** Days

### 3.2.4 Readmission Rate

**Definition:** Percentage of patients who were readmitted to the hospital.

**Formula:**  $\text{Readmission Rate} = \text{SUM}([\text{Readmission Flag}]) / \text{COUNT}([\text{Patient ID}])$

**Purpose:** Provides an indication of patient outcomes and potential follow-up care requirements.

**Measurement:** Percentage

### 3.2.5 Bed Utilization Rate

**Definition:** Percentage of available beds that are being utilized.

**Formula:**  $\text{Bed Utilization Rate} = \text{SUM}([\text{Beds Occupied Count}]) / \text{SUM}([\text{Dept Bed Capacity Derived}]) * 100$

**Purpose:** Measures the utilization of hospital bed capacity.

**Measurement:** Percentage

### 3.2.6 Department Efficiency Score

**Definition:** A composite measure used to compare the operational performance of different hospital departments. The score was derived using relevant departmental performance and resource-utilization measures.

**Formula:**  $(\text{AVERAGE}([\text{Staff Utilization \% Derived}]) + \text{AVERAGE}([\text{Bed Occupancy Rate \%}])) / 2$

**Purpose:** Allows departments to be compared based on their operational efficiency.

**Measurement:** Score

### 3.2.7 Staff Utilization Rate

**Definition:** Measures staff utilization relative to the number of occupied beds.

**Formula:**  $\text{Staff Utilization Rate} = ([\text{Beds Occupied Count}] / [\text{Staff Count}]) * 100$

**Purpose:** Provides an indication of the relationship between patient load and available staff resources.

**Measurement:** Percentage

### 3.2.8 Equipment Availability Rate

**Definition:** Percentage of required or available equipment that is currently available for use.

**Formula:** Equipment Availability Rate = COUNT( IF [Equipment Status] = "Available" THEN [Equipment ID] ) / COUNT([Equipment ID])

**Purpose:** Helps monitor the availability of hospital equipment and identify potential resource constraints.

**Measurement:** Percentage

### 3.2.9 Total Patient Transfers

**Definition:** Total number of patients who were transferred.

**Formula:** Total Patient Transfers = DISTINCTCOUNT( IF [Transfer To Department] <> "Not Transferred" THEN [Patient ID] )

**Purpose:** Measures the volume of patient movement between departments or hospital units.

**Measurement:** Count

### 3.2.10 Transfer Rate

**Definition:** Percentage of patients who were transferred.

**Formula:** Transfer Rate = DISTINCTCOUNT( IF [Transfer To Department] <> "Not Transferred" THEN [Patient ID] ) / DISTINCTCOUNT([Patient ID])

**Purpose:** Helps analyze patient movement and transfer patterns within the hospital.

**Measurement:** Percentage

### 3.2.11 Total Number of Departments

**Value:** 10

## 3.3 KPI Results

The calculated KPIs were integrated into Power BI visualizations and KPI cards.

The results can be dynamically updated based on dashboard filters such as:

- Year
- State
- Hospital
- Hospital Type
- Department
- Other dashboard-level selections

Therefore, KPI values may change when users interact with the dashboard.

## 4. Dashboard Development

---

### 4.1 Overview

The dashboard development stage focused on transforming the engineered healthcare KPIs into interactive and visually understandable dashboards using Power BI.

Four dashboards were developed to provide different perspectives of hospital operations:

- Hospital Overview Dashboard
- Patient Flow Dashboard
- Department Analytics Dashboard
- Resource Utilization Dashboard

The dashboards were designed with interactive visualizations, KPI cards, slicers/filters, and dashboard navigation buttons to help users explore hospital performance and identify important trends.

4.2 Technology Used

| Technology                | Purpose                                 |
|---------------------------|-----------------------------------------|
| Power BI Desktop          | Dashboard development and visualization |
| Excel / CSV               | Data source                             |
| Python (Jupyter Notebook) | Data cleaning and transformation        |
| GitHub                    | Project and documentation management    |

4.3 Hospital Overview Dashboard

Purpose

The Hospital Overview Dashboard provides a high-level summary of hospital performance and allows users to quickly understand overall operational conditions.

| Business Question                                        | Answer                                                                | Chart Type           |
|----------------------------------------------------------|-----------------------------------------------------------------------|----------------------|
| What is the total number of admissions?                  | 10,000                                                                | KPI Card             |
| What is the current bed occupancy rate?                  | 36.8%                                                                 | KPI Card             |
| What is the readmission rate?                            | 49.9%                                                                 | KPI Card             |
| Which admission type contributes most admissions?        | Urgent Type                                                           | Donut Chart          |
| How are admissions changing every month?                 | The trend shows monthly increases or decreases in patient admissions. | Line Chart           |
| How is occupancy changing every month?                   | The trend shows monthly variations in hospital bed occupancy.         | Line Chart           |
| Is readmission rate increasing or decreasing over time?  | The trend shows monthly changes in the hospital readmission rate.     | Line Chart           |
| Which hospitals have the highest readmission percentage? | Deepak Hospital                                                       | Horizontal Bar Chart |



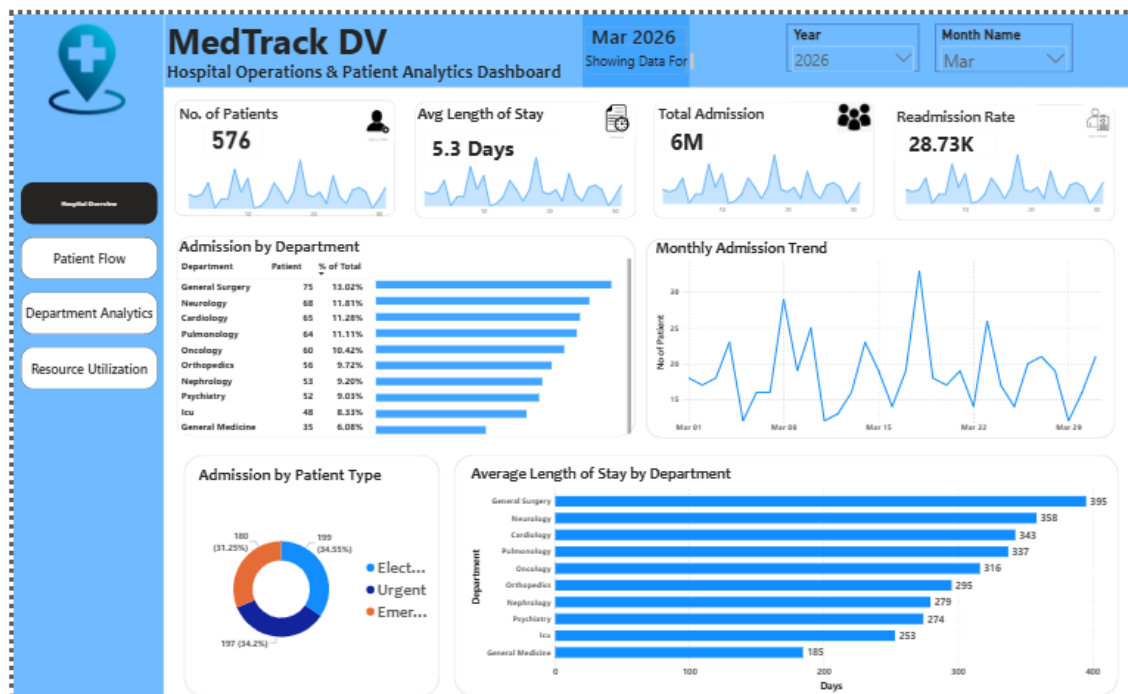


Figure 2: Hospital Overview Dashboard

## 4.4 Patient Flow Dashboard

### Purpose

The Patient Flow Dashboard focuses on the movement of patients through the hospital, from admission to transfer and other stages of their hospital journey.

| Business Question                                                    | Answer                                                        | Chart Type           |
|----------------------------------------------------------------------|---------------------------------------------------------------|----------------------|
| What is the Average Length of Stay across the hospital?              | 78.78 days                                                    | KPI Card             |
| Which months experience the highest and lowest patient admissions?   | January - Highest, November - Lowest                          | Bar Chart            |
| How have patient discharges changed over time?                       | The trend shows monthly changes in the patient discharges.    | Area Chart           |
| Which departments transfer the most patients?                        | Neurology                                                     | Horizontal Bar Chart |
| What are the most common patient movement paths between departments? | Nephrology to Oncology                                        | Heat Map (Matrix)    |
| How has the average length of stay changed over time?                | The trend shows monthly variations in average length of stay. | Line Chart           |
| What is the transfer rate of patients?                               | 17.6%                                                         | KPI Card             |
| What is the total number of patient transfers?                       | 1,760                                                         | KPI Card             |

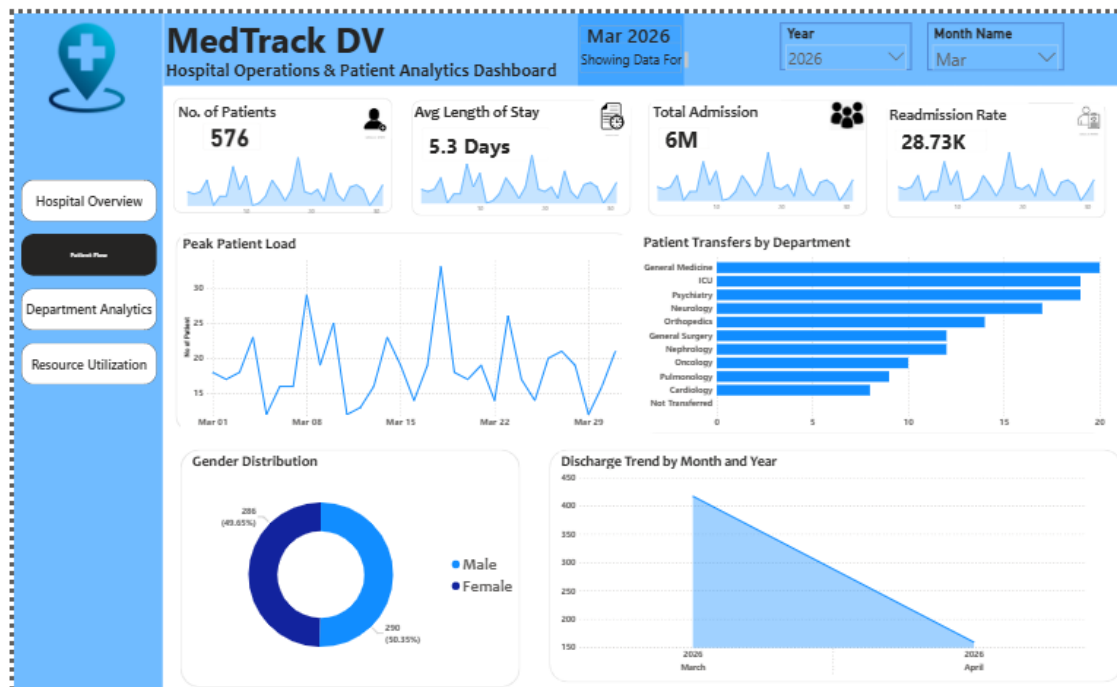


Figure 3: Patient Flow Dashboard

## 4.5 Department Analytics Dashboard

### Purpose

The Department Analytics Dashboard evaluates the performance of individual hospital departments.

| Business Question                                                                       | Answer                                                                    | Chart Type           |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------|
| What is the department efficiency score?                                                | 29.15                                                                     | KPI Card             |
| Which departments have the highest and lowest Department Efficiency scores?             | ICU - Highest, Nephrology - Lowest                                        | Horizontal Bar Chart |
| Which departments have the shortest and longest Average Length of Stay (ALOS)?          | Oncology - Longest, Cardiology - Shortest                                 | Horizontal Bar Chart |
| How does monthly patient volume vary across departments?                                | The trend shows monthly variations in patient volumes across departments. | Line Chart           |
| Which departments have the highest and lowest bed occupancy relative to their capacity? | Psychiatry - Highest, General Medicine - Lowest                           | Horizontal Bar Chart |
| Which department has the highest readmission rate?                                      | Oncology                                                                  | Horizontal Bar Chart |
| What is the transfer rate of patients?                                                  | 17.6%                                                                     | KPI Card             |
| What is the total number of departments?                                                | 10                                                                        | KPI Card             |

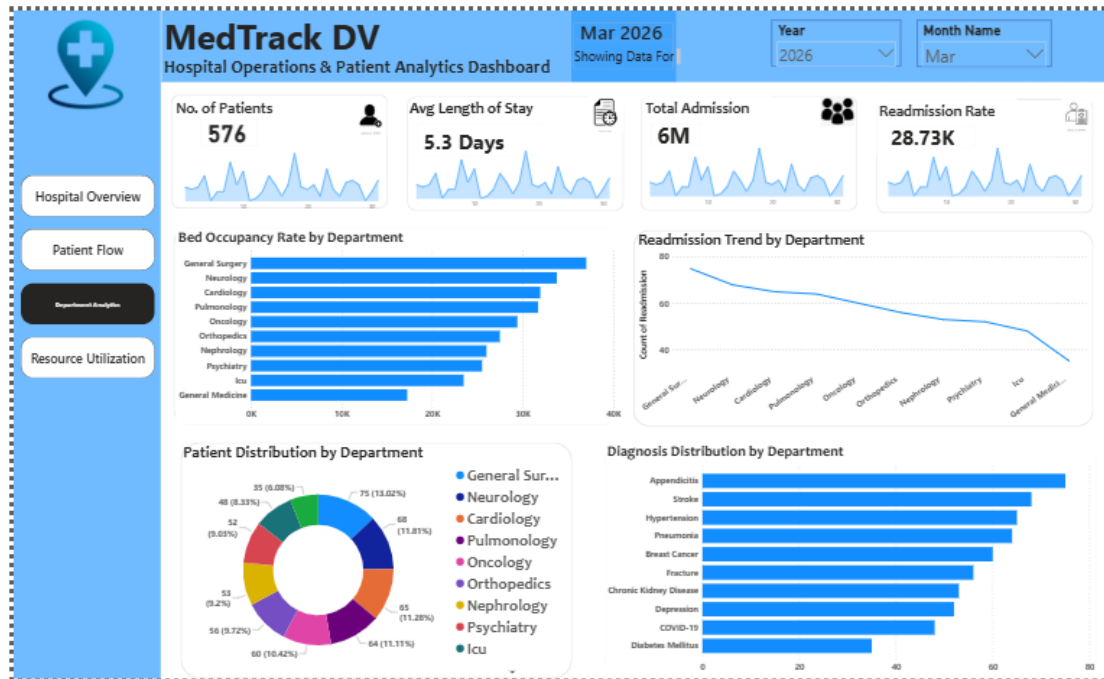


Figure 4: Department Analytics Dashboard

## 4.6 Resource Utilization Dashboard

### Purpose

The Resource Utilization Dashboard focuses on the utilization and availability of hospital resources such as beds, staff, and equipment.

| Business Question                                                              | Answer                                                 | Chart Type           |
|--------------------------------------------------------------------------------|--------------------------------------------------------|----------------------|
| What is the bed occupancy rate?                                                | 37.3%                                                  | KPI Card             |
| Which hospitals have the highest number of occupied beds?                      | Medindia Hospital                                      | Horizontal Bar Chart |
| What is the equipment availability rate?                                       | 32.9%                                                  | KPI Card             |
| Which departments have the highest bed capacity?                               | Neurology                                              | Bar Chart            |
| Which departments have the highest staff utilization percentage?               | Nephrology                                             | Bar Chart            |
| Which equipment types are used for the longest duration?                       | Patient Monitor                                        | Bar Chart            |
| What is the distribution of equipment status (Available, In Use, Maintenance)? | Available - 32.9%, Maintenance - 33.6%, In Use - 33.6% | Donut Chart          |
| What is the staff utilization rate?                                            | 70.8%                                                  | KPI Card             |

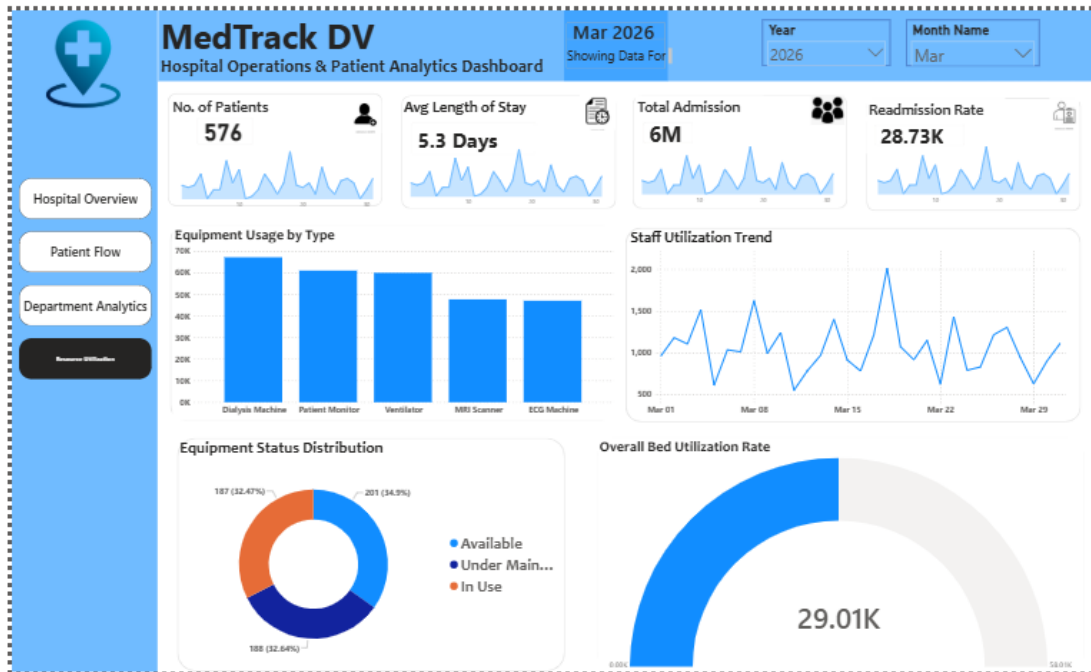


Figure 5: Resource Utilization Dashboard

## 5. Dashboard Integration

Interactive functionality was incorporated to make the dashboards dynamic and user-friendly. The following features were implemented:

**Slicers & Filters:** Slicers allow users to view data for specific years, states, hospitals, departments, and other relevant dimensions.

**Dashboard Actions (Cross-Filtering):** Filter interactions and cross-highlighting were used to enable interaction between visualizations and filters.

**Navigation Controls:** Navigation buttons (built using bookmarks and button actions) were incorporated to allow users to move between the different dashboards.

**Dashboard Linking:** The dashboards were connected through navigation buttons, allowing users to move between different analytical perspectives.

## 6. Testing & Validation

### 6.1 Overview

The testing and validation stage was performed to ensure that the cleaned dataset, calculated KPIs, Power BI visualizations, slicers, dashboard actions, and navigation controls worked correctly.

The objective was to verify the accuracy, consistency, functionality, and usability of the final healthcare analytics dashboard.

### 6.2 QA Checklist

A detailed, itemized QA Checklist covering KPI validation, healthcare metrics, patient-flow analytics, and dashboard interaction testing is provided as a separate document (QA\_Check\_List). A summary is provided below:

| Test Category                     | Status |
|-----------------------------------|--------|
| Data validation                   | Passed |
| KPI validation                    | Passed |
| Dashboard visualization testing   | Passed |
| Dashboard-action / filter testing | Passed |
| Navigation controls               | Passed |
| Dashboard linking                 | Passed |
| Visual/layout validation          | Passed |

## 7. User Guide

---

### User Workflow

Open Dashboard → Select Required Filters → Review KPIs → Interact with Visualizations → Navigate to Related Dashboard → Analyze Insights → Clear Filters for Overall View

## 8. Conclusion

---

The Hospital Analytics Dashboard project successfully transformed raw hospital data into a structured and interactive analytics solution. The project followed a systematic workflow covering data collection, data cleaning and transformation, healthcare KPI engineering, dashboard development, dashboard integration, testing, validation, and final documentation.

The hospital dataset was cleaned and prepared to improve data quality and consistency. Important healthcare KPIs such as Total Admissions, Occupancy Rate, Average Length of Stay, Readmission Rate, Bed Utilization Rate, and Department Efficiency Score were developed to measure hospital performance.

Four interactive Power BI dashboards were developed: Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization. These dashboards allow users to analyze hospital performance, patient movement, department-level operations, and resource utilization through interactive visualizations and filters.

Dashboard integration features such as slicers, filters, navigation controls, dashboard linking, and button actions improved the overall usability and interactivity of the solution. Testing and validation were also performed to ensure that KPIs, filters, visualizations, and dashboard actions function correctly.

Overall, the project demonstrates how data analytics and visualization can convert complex healthcare data into meaningful insights. The completed dashboard provides a user-friendly platform for monitoring hospital operations and supports data-driven decision-making and performance analysis.